

Notas e Listas de Prob 2

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October 6, 2026

Contents

0	Informações importantes	1
1	Listas	1
1.1	Lista 1	1
2	Lista 2	4

0 Informações importantes

1 Listas

1.1 Lista 1

Vou passar a limpo exercícios do Varadhan.

Exercise 1.1 (5.1). Construct a nonnegative martingale X_n with $\mathbb{E}X_n = 1$ such that $\xi = \sup_n X_n$ is not integrable.

Proof. The proof is basically done in the book. Just consider the Lebesgue measure on $[0, 1]$ and the $\mathcal{F}_n$ will be the sigma-algebras generated by the diadic sets of length $1/2^n$. Then define

$$X_n = 2^n \mathbf{1}[\omega \in [0, 2^{-n}]].$$

It is easy to check it is a martingale. Furthermore,

$$\int_{[0,1]} \xi dx = \sum_{j=0}^{\infty} \int_{[2^{-j-1}, 2^{-j}]} X_j dx = \sum_{j=0}^{\infty} 1/2 = \infty.$$

□

Exercise 1.2 (5.2). If X_n is a martingale such that the differences $Y_n = X_n - X_{n-1}$ are all square integrable, show that for $n \neq m$, $\mathbb{E}[Y_n Y_m] = 0$. Therefore

$$\mathbb{E}[X_n^2] = \mathbb{E}[X_0^2] + \sum_{j=1}^n \mathbb{E}[Y_j^2].$$

If in addition, $\sup_n \mathbb{E}[X_n^2] < \infty$, then show there is a random variable X such that

$$\lim_{n \rightarrow \infty} \mathbb{E}[|X - X_n|^2] = 0$$

Proof. Without loss of generality, suppose $n > m$. Then, by simply conditioning on $\mathcal{F}_{n-1}$

$$\mathbb{E}[Y_n Y_m] = \mathbb{E}[(X_n - X_{n-1})Y_m] = \mathbb{E}[Y_m(\mathbb{E}[X_n | \mathcal{F}_{n-1}] - X_{n-1})] = 0.$$

The other equality becomes obvious from the fact that $X_n = X_0 + \sum_{j=1}^n Y_j$. For the conclusion, note that together with the previous observation, it implies

$$\sum_n \mathbb{E}Y_n^2 < \infty$$

so that the X_n form a Cauchy sequence in L^2 . As L^2 is complete, there exists such X . □

Exercise 1.3 (5.3). Show there exists a uniformly bounded in L^1 martingale X_n but X_n does not converge in L^1 .

Proof. The first exercise provides an example. It is clear that $X_n \rightarrow 0$ almost surely, but $|X_n|_{L^1} = 1$. This is an absurd, since any limit of X_n in L^1 would have to be a.s equal to $\lim_{n \rightarrow \infty} X_n$. □

Exercise 1.4 (5.4). Establish that a nonnegative martingale has an almost sure limit (which may not be in L^1). Show (I won't really do it) that we can assume, wlog that we are in the following situation:

$$\Omega = \bigotimes_{j=1}^{\infty} \mathbb{R}, \quad \mathcal{F} = \sigma(x_1, \dots, x_n), \quad X_j(\omega) = x_j.$$

The existence of a Q such that

$$\frac{dQ}{dP}|_{\mathcal{F}_n} = x_n$$

is essentially a Kolmogorov's consistency theorem.

Proof. Actually don't know how to do this one. □

Exercise 1.5 (5.5). Show that $\tau(\omega) \equiv k$ is a stopping time for any constant k .

Proof. Note that if $n < k$, then $\{\tau \leq n\} = \emptyset \in \mathcal{F}_n$. Similarly, if $n \geq k$, $\{\tau \leq n\} = \Omega \in \mathcal{F}_n$. □

Exercise 1.6 (5.6). Show that if τ is a stopping time and $f : \mathbb{N} \rightarrow \mathbb{N}$ is a non-decreasing function with $f(t) \geq t$ for all t . Then $f(\tau)$ is also a stopping time.

Proof. Just note that given n , there exists a maximal m s.t $f(m) \leq n$ and by hypothesis, $m \leq n$. Then $\{f(\tau) \leq n\} = \{\tau \leq m\} \in \mathcal{F}_m \subset \mathcal{F}_n$. □

Exercise 1.7 (5.7). Show that if τ_1 and τ_2 are stopping times, so is $\max(\tau_1, \tau_2)$ and $\min(\tau_1, \tau_2)$. In particular, τ is an increasing limit of bounded stopping times $\tau \wedge n$.

Proof. Fix n , note that $\{\tau_1 \wedge \tau_2 \leq n\} = \{\tau_1 \leq n\} \cup \{\tau_1 \leq n\} \in \mathcal{F}_n$. The same is true for the max, we just take intersection instead of union. $\square$

Exercise 1.8 (5.8). Verify that for a stopping time τ , $\mathcal{F}_\tau$ is a sub σ -field. If $\tau \equiv k$, then $\mathcal{F}_\tau = \mathcal{F}_k$. If $\tau_1 \leq \tau_2$ are stopping times, then $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$. Finally, if τ is a stopping time, then it is $\mathcal{F}_\tau$ measurable.

Proof. Let's show that $\mathcal{F}_\tau \subset \mathcal{F}$ is a σ -field. Clearly $\emptyset \in \mathcal{F}_{\{\tau\}}$. Let $A \in \mathcal{F}_\tau$, then for every n , $A \cap \{\tau \leq n\} \in \mathcal{F}_n$. But then, $A^c \cup \{\tau > n\} \in \mathcal{F}_n$, and therefore $A^c \cap \{\tau \leq n\} \in \mathcal{F}_n$. Similarly, if $(A_n)_{n=1}^\infty$ is a sequence of sets in $\mathcal{F}_\tau$, then for any m

$$\{\tau \leq m\} \cap \bigcup_{n \geq 1} A_n = \bigcup_{n \geq 1} \{\tau \leq m\} \cap A_n \in \mathcal{F}_m.$$

If $\tau \equiv k$. It is clear that $\mathcal{F}_k \subset \mathcal{F}_\tau$ since for any $A \in \mathcal{F}_k$, $A \cap \{\tau \leq k\} = A \in \mathcal{F}_k$. Similarly, if $A \in \mathcal{F}_\tau$, then $A = A \cap \tau = k \in \mathcal{F}_k$.

If $\tau_1 \leq \tau_2$ and $A \in \mathcal{F}_{\tau_1}$ then for any m $A \cap \{\tau_2 \leq m\} \subset (A \cap \{\tau_1 \leq m\}) \cap \{\tau_2 \leq m\}$. Note that $A \cap \{\tau_1 \leq m\} \in \mathcal{F}_m$ and as $\{\tau \leq m\} \in \mathcal{F}_m$, we have that their intersection is also in $\mathcal{F}_m$.

Finally, measurability is obvious since $\{\tau \leq n\} \in \mathcal{F}_\tau$ for every n . $\square$

Exercise 1.9 (5.10). If X_n is a submartingale and $\tau_1 \leq \tau_2 \leq M$ are bounded stopping times, we have that $\mathbb{E}[X_{\tau_2} | \mathcal{F}_{\tau_1}] \geq X_{\tau_1}$. With the respective result for supermartingales.

Proof. Decompose X_n as $Y_n + A_n$ where Y_n is a martingale and A_n is strictly increasing and predictable. Now it is clear that, by linearity,

$$\mathbb{E}[Y_{\tau_2} + A_{\tau_2} | \mathcal{F}_{\tau_1}] = Y_{\tau_1} + \mathbb{E}[A_{\tau_2} | \mathcal{F}_{\tau_1}] \geq Y_{\tau_1} + A_{\tau_1}.$$

since $A_{\tau_2} \geq A_{\tau_1}$.

The interesting case is when X_n is a martingale. Then you may only assume $\tau \leq M$ since

$$\mathbb{E}[X_m | \mathcal{F}_{\tau_1}] = \mathbb{E}[\mathbb{E}[X_m | \mathcal{F}_{\tau_2}] | \mathcal{F}_{\tau_1}].$$

Then we are left with proving that $\mathbb{E}[X_M | \mathcal{F}_\tau] = X_\tau$, to do that just separate on the values of τ . $\square$

Exercise 1.10 (5.11). Let $X_0 = 0$ and $X_n = \sum_{j \leq n} \xi_j$ where ξ_j are u.i.i.d on $\{-1, 1\}$. Let $\tau = \inf\{n : X_n = 1\}$. Then τ is a stopping time, $\mathbb{P}(\tau < \infty) = 1$, but τ is not bounded. $X_\tau = 1$ with probability 1 but $\mathbb{E}[X_0] = 0$.

Proof. I think the only thing to show on this exercise is that τ is a stopping time and $\mathbb{P}(\tau < \infty) = 1$. Note that $\{\tau = n\} = \{X_n = 1\} \cap \bigcap_{j \leq n} \{X_j \neq 1\} \in \mathcal{F}_n$, so it is a stopping time. To show recurrence we can proceed with the usual Catalan counting or use markov chains arguments. $\square$

Exercise 1.11 (5.12). Let τ be an unbounded stopping time. For every n , $\tau \wedge n$ is a bounded stopping time and $\mathbb{E}X_{\tau \wedge n} = 0$. As $n \rightarrow \infty$, $\tau \wedge n \rightarrow \tau$. If we can establish uniform integrability, then we can pass to the limit. In particular, is $S(\omega) = \sup_{0 \leq n \leq \tau} |X_n(\omega)|$ is integrable, then $\sup_n |X_{\tau \wedge n}| < S$ and so $\mathbb{E}X_\tau = 0$.

Exercise 1.12. Seja τ um stopping time e $\tau_k \rightarrow \tau$ uma sequência crescente de stopping times. Mostre que

$$\mathcal{F}_\tau = \sigma\left(\bigcup_k \mathcal{F}_{\tau_k}\right)$$

Precisamos de algumas observações simples.

Lemma 1.1. Sejam $\tau_1 \leq \tau_2$ stopping times. Então $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$.

Proof. Tome $A \in \mathcal{F}_{\tau_1}$. Por definição, para todo n temos $A \cap \{\tau_1 \leq n\} \in \mathcal{F}_n$. Como $\tau_1 \leq \tau_2$ e $\{\tau_2 \leq n\} \in \mathcal{F}_n$,

$$A \cap \{\tau_2 \leq n\} = (A \cap \{\tau_1 \leq n\}) \cap \{\tau_2 \leq n\} \in \mathcal{F}_n.$$

Como isso vale para todo n , $A \in \mathcal{F}_{\tau_2}$. □

Observation 1.2. Seja $B \in \mathcal{F}_n$ e τ um stopping time qualquer. Então vale que

$$B \cap \{\tau = n\} \in \mathcal{F}_\tau \quad \& \quad B \cap \{\tau > n\} \in \mathcal{F}_\tau.$$

Prova do Exercício. Segue imediatamente do primeiro lema que $\sigma(\bigcup_k \mathcal{F}_{\tau_k}) \subset \mathcal{F}_\tau$. Basta verificar a outra inclusão. Seja $A \in \mathcal{F}_\tau$, separamos A de acordo com o que acontece em τ

$$A = A \cap \{\tau = \infty\} \cup \left(\bigcup_n A \cap \{\tau \leq n\} \right).$$

Vamos provar que cada pedacinho pertence a $\sigma(\bigcup_k \mathcal{F}_{\tau_k})$.

Por definição, $A \cap \{\tau \leq n\} \in \mathcal{F}_n$. Como $\tau_k \rightarrow \tau$, segue que

$$A \cap \{\tau \leq n\} = \bigcup_k [A \cap \{\tau \leq n\} \cap \{\tau_k = n\}].$$

Pela observação, anterior, $A \cap \{\tau \leq n\} \cap \{\tau_k = n\} \in \mathcal{F}_{\tau_k}$. Sendo união de conjuntos dessa forma, $A \cap \{\tau \leq n\} \in \sigma(\bigcup_k \mathcal{F}_{\tau_k})$. Para $\{\tau = \infty\}$ temos basicamente a mesma coisa. Note que

$$A \cap \{\tau = \infty\} = \bigcap_n A \cap \{\tau > n\}.$$

Usando que $\tau_k \rightarrow \tau$, $A \cap \{\tau > n\} = \bigcup_k A \cap \{\tau > n\} \cap \{\tau_k > n\}$. Da mesma forma que antes, cada um dos termos pertence a $\mathcal{F}_{\tau_k}$ e o resultado segue. □

2 Lista 2

Alguns exercícios do Durrett de Markov Chains.